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The Editor-in-Chief
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Dear Editor-in-Chief,

We respectfully submit our manuscript, “**NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction**,” for consideration as a Research Paper in *Signal Processing*, an international journal of the European Association for Signal Processing.

Scope Fitness Assessment. NARCIS addresses image-based communication through unchanged natural images. It combines self-supervised representation learning, scalar projection and quantile partitioning, attack-qualified signal classification, finite-index feasibility, and channel coding. The work lies at the intersection of machine learning for signal processing, image and multimedia signal processing, detection and estimation, and signal processing for communications.

- **Signal representation:** a compact neural descriptor is trained to preserve image labels under declared channel distortions.
- **Detection and communication:** an alphabet is accepted only when every coded symbol has sufficient qualified images in the finite index.
- **Evaluation:** calibration and holdout distortions are separated; recovery, selection bias, complexity, and payload overhead are measured.

Scientific Contribution and Evidence. Across five deterministic partitions each of BOSSBase and RGB Caltech-101, all 1,575 clean, calibration, and holdout message-condition trials recovered the authenticated plaintext. Mean symbol accuracies were 98.74% and 98.59%. A matched BOSSBase ablation fell from 210/210 to 142/210 recoveries without qualification. At a 16-symbol operating point, net rate increased from 0.184 to 1.213 bits/cover for 8- to 128-byte payloads. Calibration-locked projection selection increased the minimum stable bucket from 92 to 161 on the evaluated BOSSBase partition. The paper also reports a weak Caltech-101 selection signal (repeated CNN AUC 0.533, 95% CI [0.517, 0.548]) and makes no general undetectability claim.

Positioning. Unlike work centred on nominal hash length or isolated descriptor accuracy, NARCIS tests whether a complete coded message can be realised by a finite image index and recovered after declared signal distortions. Its position is joint representation, classification, channel coding, and verified operating-point selection, not universally maximal capacity.

Mandatory Author Declarations.

- **Originality/exclusivity:** The manuscript is original, unpublished, and not under consideration elsewhere.
- **Author approval:** All authors approved the manuscript, order, contributions, and submission.
- **Interests/funding:** No competing interests or external funding are declared.
- **Ethics/reproducibility:** No humans, animals, or personal data are involved. Code, seeds, evidence, tests, and sources are available at github.com/EkodeckStephane/NARCIS.

The manuscript fits the journal’s readership at the intersection of machine learning for signal processing, robust image communication, multimedia processing, and reproducible algorithm evaluation. Thank you for considering this submission.

Yours sincerely,

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